

EDUCATION

Seoul National University (SNU)

Bachelor of Science in Mechanical Engineering

Minor in Philosophy

Seoul, South Korea

Expected 08/2026

TOEFL iBT : 105 / 120

GPA : 3.76 / 4.00

PUBLICATIONS

SPREAD: Scalable Pre-trained World Model for Adaptive Dynamics Model

Jihun Moon, Seong-Woo Kim

Under Review at RA-L 2026

[Paper Link](#)

Submitted 11/2025

Beyond Destinations: Instruction-Aware Graph Path Planning and Navigation with OpenStreetMap

Donghwi Jung, Jihun Moon, Jiyang Lee, DongYeop Shin, Jungjae Lee, Jinhee Kim, Seong-Woo Kim

Under Review at ICRA 2026

[Paper Link](#)

Submitted 09/2025

RESEARCH EXPERIENCE

Autonomous Robot Intelligence Laboratory, SNU

Undergraduate Researcher

Advisor: Prof. Kim, Seong woo

Seoul, South Korea

05/2025 – Present

- **Independent Research on World Model:** Proposed online continual learning method for pre-trained world model architecture, designed for real-time adaptation to diverse environmental changes in continual robot learning
- **Language-Guided Navigation Planning:** Designed planning framework using LLM and VLM to extract destination, route constraints, and avoidances from user instructions, and to generate and refine paths
- **Language-Vision World Model:** Exploring how natural language feedback can be incorporated into language-vision world models

Transformative Architecture Laboratory, SNU

Undergraduate Researcher

Advisor: Prof. Yang, Jinkyu

Seoul, South Korea

08/2024 – 11/2024

- **Origami Actuator Fabrication:** Fabricated multi-DOF origami actuator capable of simultaneous folding and unfolding through 3D CAD modeling and structural design optimization

EXTRACURRICULAR ACTIVITIES

Humanoids Club 'SNU Humanoid Club'

Founding President

Advisor: Prof. Kim, Seong woo

SNU

11/2025 – Present

- **Organization Founding:** Established Seoul National University's first student organization dedicated to humanoid robotics as a founding president, focused on full-scale humanoid development
- **Humanoid Development:** Leading hands-on operation of humanoid robots and representing the club in international home-service humanoid competitions, emphasizing autonomous task execution from natural language instructions
- **Club Management:** Managing club administration, including recruitment, project planning, and team operations, to build and sustain a multidisciplinary student research community

Robotics Club 'SIGMA'

Member

SNU

03/2024 – 06/2024

- Prototyped line-tracing robot by applying motor control using Arduino; participated in Linux and Raspberry Pi workshop

Philosophy Academic Club 'Hapum'

Member

SNU

09/2024 – 12/2024

- Discussed analytic philosophy, focusing on Dummett's and Putnam's arguments on realism

MILITARY SERVICE

Republic of Korea Army, Joint Security Area Battalion (JSA, United Nations Command)

Paju, South Korea

Sergeant

12/2021 – 06/2023

- Served at South Korea's northernmost front-line outpost, conducting surveillance and security operations near the North Korean border

SCHOLARSHIPS

- | | |
|---|-----------------|
| ● SNU Merit-based Scholarship for Academic Excellence | Awarded 08/2024 |
| ● SNU ME Additional school Scholarship | Awarded 12/2024 |
| ● SNU Merit-based Scholarship for Academic Excellence | Awarded 02/2025 |

SELECTED PROJECTS

Maze Solving Manipulation

Project for Course 'Dynamics and Control of Robot-Environment Interaction'

10/2025 – 12/2025

- Implemented an A*-based path planner with a feedback trajectory tracking controller; achieved the top performance among all teams

Automatic Book Scanner

11/2024 – 12/2024

Project for Course 'Mechatronics'

- Designed and fabricated a multi-motor control system for automatic book scanner capable of page turning and synchronized screen transitions using a film suction pad mechanism

Autonomous Mini Vehicle

10/2021 – 12/2021

Project for Course 'Creative Engineering Design'

- Designed sensor-based mini vehicle operating through linkage mechanism instead of wheels; achieved 2nd place among all teams

SKILLS

Machine Learning & AI

- **Pytorch:** Enhanced ViT-based world model and Implemented LoRA-based finetuning
- **LLM/VLM:** Used LLMs and VLMs for instruction parsing, constraint extraction, and prompt design

Robot Control & Dynamics

- **Model-based Control:** Implemented model predictive control integrated with learned world model
- **Control Theory:** Completed graduate course "*Dynamics and Control of Robot-Environment Interaction*"

ROS & MuJoCo

- **ROS 2:** Conducted high-level path planning and navigation for ROS-based quadruped robot project
- **MuJoCo:** Utilized control theory to MuJoCo-based manipulator control and motion planning

Programming

- **Python, Linux:** Used extensively in Python for both team and independent research projects